

SVM (Support Vector Machine)

↳ SVM is one of most popular Supervised Learning algorithms.

↳ It is used for Classification as well as Regression problems.

↳ It is primarily used for classification problem in machine learning.

↳ SVM is a powerful machine learning algorithm used for linear or nonlinear classification, regression, and even outlier detection tasks.

↳ The goal of the SVM algorithm is to create the best line or decision boundary that can segregate n -dimensional space into classes so that we can easily put the new data point in correct category in the future. This best decision boundary is called a hyperplane.

↳ SVM chooses the ~~extrem~~ extreme points / vectors that help in creating the hyperplane. These extreme cases are called as support vectors and hence algorithm is termed as Support Vector Machine.

↳ The hyperplane tries that the margin between the closest points of different classes should be as maximum as possible.

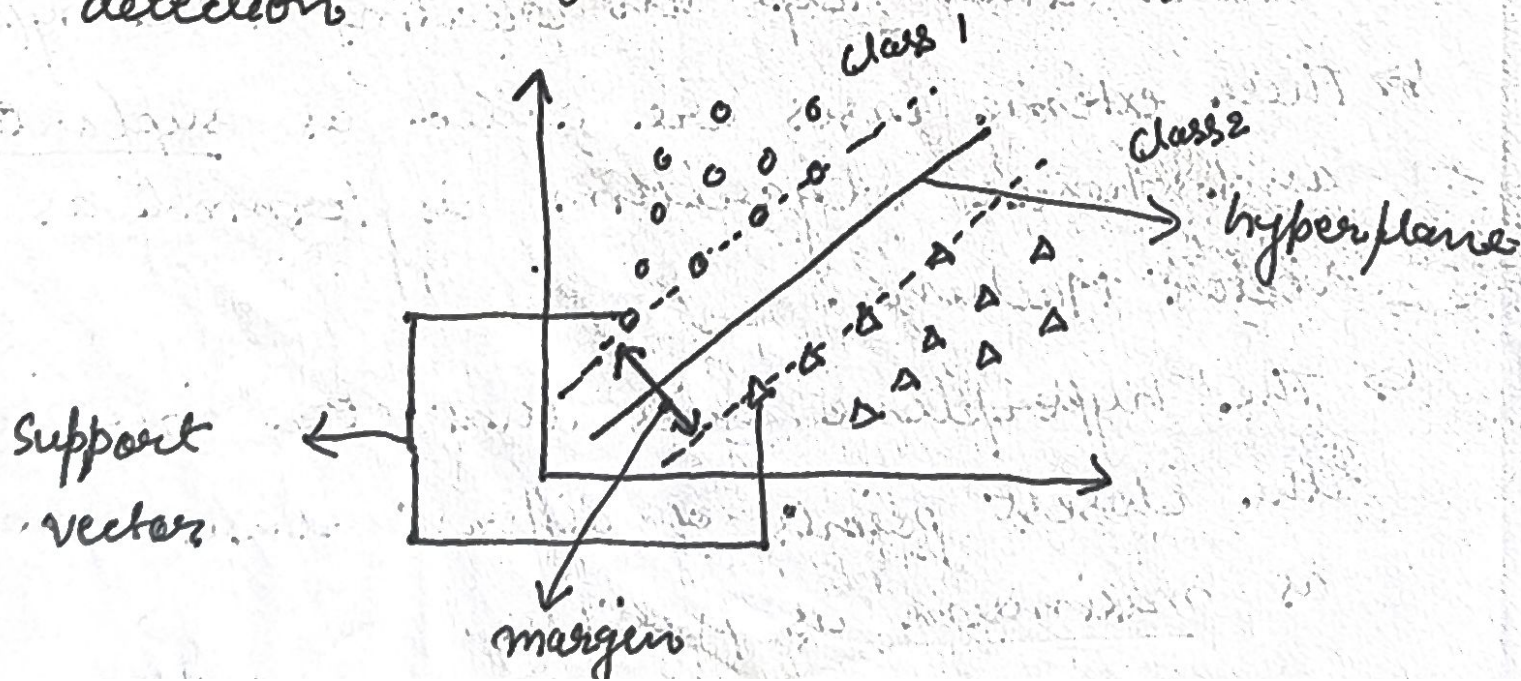
↳ To separate the two classes of data points, there are many possible hyperplane that could be chosen.

↳ Our objective is to find a plane that has the maximum margin, i.e. the maximum distance between data points of both classes.

↳ Maximizing the margin distance provides some reinforcement so that future data points can be classified with more confidence.

↳ The dimension of the hyperplane depends upon the number of features. If the number of fp feature is two, then the hyperplane is just a line. If the number of fp features is three, then the hyperplane becomes a 2.D plane.

↳ SVM can be used for a variety of tasks such as text classification, image classification, spam detection, handwriting identification, gender expression analysis, face detection and anomaly detection.



Support Vector Machine:

Optimal Hyperplane for Linearly Separable Pattern

Consider training sample $\{(x_i, d_i)\}_{i=1}^N$

where x_i = input pattern
 d_i = target output.

Assume $d_i = +1$ & $d_i = -1$ classes are linearly separable. then equation of a decision surface in the form of hyperplane as below

$$w^T x + b = 0 \quad \text{--- (1)}$$

where x is an i/p vector

w is an adjustable weight vector
 b is a bias

we may write

$$w^T x_i + b \geq 0 \quad \text{for } d_i = +1$$

$$w^T x_i + b < 0 \quad \text{for } d_i = -1 \quad \text{--- (2)}$$

→ The separation between the hyperplane and the closest data point is called the margin of separation, denoted by ρ .

→ The goal of SVM is to find the particular hyperplane for which the margin of separation (ρ) is maximized. Under this condition, the decision surface is referred to as the optimal hyperplane.

→ Let w_0 & b_0 denote the optimal value then optimal hyperplane will be

$$w_0^T x + b_0 = 0 \quad \text{--- (3)}$$

rewritten

$$g(x) = w_0^T x + b_0 \quad \text{--- (4)}$$

Distance from x to optimal hyper plane

$$L = \frac{g(x)}{\|w_0\|} \quad \text{--- (5)}$$

→ Distance from origin (i.e. $x=0$) to optimal hyper plane is given by $\frac{b_0}{\|w_0\|}$ (putting $x=0$ in equation)

• ↳ if $b_0 > 0$, the origin is on the positive side of optimal parameter hyperplane

• ↳ if $b_0 < 0$, the origin is on the negative side of optimal hyperplane.

• ↳ if $b = 0$, the optimal hyperplane passes through the origin.

To find the parameters w_0 and b_0 for the optimal hyperplane, given the training set S of $\{(x_i, d_i)\}$, then (w_0, b_0) must satisfy the following constraint.

$$w_0^T x_i + b_0 \geq 1 \quad \text{for } d_i = +1$$

$$w_0^T x_i + b_0 \leq -1 \quad \text{for } d_i = -1$$

--- (6)

The particular data point (x_i, d_i) for which first or second line of equation of equation (6) is satisfied with the equality sign are called support vectors. These support vectors are those data points that lie closest to the optimal hyperplane and are therefore the most difficult to classify.

for support vector $x^{(s)}$ for which $d^{(s)} = \pm 1$, then

$$g(x^{(s)}) = w_0^T x^{(s)} + b = \pm 1 \text{ for } d^{(s)} = \pm 1 \quad (7)$$

from equation (5), distance from the support vector $x^{(s)}$ to the optimal hyperplane is

$$r = \frac{g(x^{(s)})}{\|w_0\|}$$

$$\hat{r} = \begin{cases} \frac{1}{\|w_0\|} & \text{if } d^{(s)} = +1 \\ -\frac{1}{\|w_0\|} & \text{if } d^{(s)} = -1 \end{cases} \quad (8)$$

where plus sign indicates that $x^{(s)}$ lies on positive side of the optimal hyperplane and the minus sign indicates that $x^{(s)}$ lies on negative side of the optimal hyperplane.

Thus margin of separation b/w the two classes that constitute the training sample of. Then from equation (8) we get.

$$\rho = 2r = \frac{2}{\|w_0\|} \quad (9)$$

↳ Any optimization problem can be formulated into two way, primal and dual problems.

↳ First we use primal formulation for optimization algorithm, but if it does not yield any solution we go for dual optimization formulation, which is guaranteed to yield solution.

↳ SVM optimization problem is case of constrained optimization problem, and it is always preferred to use dual optimization algo. to solve such constrained optimization problems. That's why we don't use gradient descent.

↳ Since it is constrained optimization problem Lagrange multipliers are used to solve it.

↳ We will divide the complete formulation into three parts.

1. In first we will formulate SVM optimization problem mathematically
2. We will find gradient with respect to learning parameters.
3. We will find the value of parameters which minimizes $\|w\|$

finding optimal values of w and b :

Training sample $\mathcal{T} = \{x_i, d_i\}_{i=1}^N$,
from equation (6) we get.

$$d_i(w^T x_i + b) \geq 1 \text{ for } i = 1, 2, \dots, N. \quad - (10)$$

Now formally state the constrained-optimization problem as follows:

$$\cancel{d_i(w^T x_i + b) \geq 1}$$

Given the training sample $\{(x_i, d_i)\}_{i=1}^N$, find the optimum values of the weight vector w and bias b such that they satisfy the constraints

$$d_i(w^T x_i + b) \geq 1 \text{ for } i = 1, 2, \dots, N$$

and the weight vector w minimize the cost function

$$\Phi(w) = \frac{1}{2} w^T w$$

This problem is called primal problem. It is characterized as follows:

- The cost function $\Phi(w)$ is convex function of w .
- The constraints are linear in w .

→ Constrained-optimization problem can be solved by using the method of Lagrange multipliers.

$$J(w, b, \alpha) = \frac{1}{2} w^T w - \sum_{i=1}^N \alpha [d_i(w^T x_i + b) - 1] \quad - (11)$$

→ auxiliary nonnegative variable α are called Lagrange multipliers.

→ The solnⁿ to the constrained-optimization problem is determined by the saddle point of the Lagrangian function $J(w, b, \alpha)$.

→ A saddle point of a Lagrangian is a point where the roots are real, but opposite signs. This saddle point has to be minimized with respect to w and b ; it also has to be maximized with respect to α . Therefore we differentiate $J(w, b, \alpha)$ with respect to w and b and setting the result equal to zero as follows.

$$\text{Condition 1: } \frac{\partial J(w, b, \alpha)}{\partial w} = 0$$

$$\text{condition 2: } \frac{\partial J(w, b, \alpha)}{\partial b} = 0$$

Application of optimality condition 1 to the Lagrangian function of equation (11) yields the following

$$w = \sum_{i=1}^N \alpha_i d_i x_i \quad \text{--- (12)}$$

Application of optimality condition 2 to the Lagrangian function of equation (11) yields the following

$$b = \sum_{i=1}^N \alpha_i d_i = 0 \quad \text{--- (13)}$$

To postulate the dual problem for our primal problem, we first expand equation (11)

$$J(w, b, \alpha) = \frac{1}{2} w^T w - \sum_{i=1}^N \alpha_i d_i w^T x_i - b \sum_{i=1}^N \alpha_i d_i + \sum_{i=1}^N \alpha_i \quad \text{--- (14)}$$

the value of derivative of b
 Putting equation (13) in equation (15) we get.

$$J(w, b, \alpha) = \frac{1}{2} w^T w - \sum_{i=1}^N \alpha_i d_i w^T x_i + 0 + \sum_{i=1}^N \alpha_i \quad (16)$$

Now putting equation (14) in equation (16) we get

$$J(w, b, \alpha) = \frac{1}{2} \sum_{i=1}^N \sum_{j=1}^N \alpha_i \alpha_j d_i d_j x_i^T x_j - \sum_{i=1}^N \sum_{j=1}^N \alpha_i \alpha_j d_i d_j x_i^T x_j + \sum_{i=1}^N \alpha_i$$

$$Q(\alpha) = \sum_{i=1}^N \alpha_i - \frac{1}{2} \sum_{i=1}^N \sum_{j=1}^N \alpha_i \alpha_j d_i d_j x_i^T x_j \quad (17)$$

In above equation (17), we have changed the notation from $J(w, b, \alpha)$ to $Q(\alpha)$ due to transformation from primal optimization to its dual. Therefore dual problem maximizes with objective function.

$$Q(\alpha) = \sum_{i=1}^N \alpha_i - \frac{1}{2} \sum_{i=1}^N \sum_{j=1}^N \alpha_i \alpha_j d_i d_j x_i^T x_j$$

Subject to the constraints

$$(1) \sum_{i=1}^N \alpha_i d_i = 0$$

$$(2) \alpha_i \geq 0 \text{ for } i=1, 2, \dots, N$$

Equation (17) is cast entirely in terms of the training data. However the function $Q(\alpha)$ to be maximized depends only on the i/p pattern in the form of a set of dot products.

SVM with kernel:

kernel Function:

- ↳ It is method used to take data as i/p and transform it into required form of processing data.
- ↳ Kernel is used due to a set of mathematical functions used in SVM.
- ↳ Kernel function generally transforms the training set of data so that a non-linear decision surface is able to transform to a linear equation in a higher number of dimension spaces.
- ↳ It returns the inner product between two point in standard feature dimension.
- ↳ The kernel function in SVM is essential in determining the decision boundary that divides the various classes.
- ↳ The most commonly used kernel function in SVM is the Gaussian or radial basis function (RBF) kernel. The RBF kernel maps the i/p data into an infinite-dimensional feature space using a Gaussian function. This kernel function is popular because it can capture complex non-linear relationship in the data.
- ↳ Other type of kernel function that can be used in SVM include the polynomial kernel, the sigmoid kernel and Laplacian kernel.

↳ The choice of kernel function depends on the specific problem and characteristics of the data:

↳ Basically kernel methods in SVM are powerful techniques for solving classification and regression problems and they are widely used in machine learning because they can handle complex data structure and are robust to noise and outliers.

Major kernel function in SVM:

- Linear kernel
- Polynomial
- Sigmoidal RBF / Gaussian function

Linear kernel:

↳ A linear kernel is type of kernel function used in machine learning in SVM.

↳ It is the simplest and most commonly used kernel function and it defines the dot product b/w the i/p vectors in the original feature space

↳ The linear kernel can be defined as:

$$K(x, y) = x \cdot y$$

Where x and y are the i/p feature vectors. The dot product of i/p vectors is a measure of their similarity or distance in the original feature space.

↳ The degree of nonlinearity in the decision boundary is determined by the degree of the polynomial.

↳ The polynomial kernel has benefit of being able to detect both linear and non-linear correlations in data.

↳ It can be difficult to select the proper degree cannot adequately represent the underlying relationships in data.

↳ In general, the polynomial kernel is an effective tool for converting the i/p data into a higher-dimensional feature space in order to capture non-linear correlations b/w the i/p characteristics.

Gaussian (RBF) kernel:

↳ The Gaussian kernel also known as the radial basis function (RBF) kernel, is a popular kernel function used in machine learning, particularly in SVM.

↳ It is a nonlinear kernel function that maps the i/p data into a higher-dimensional feature space using a Gaussian function.

↳ The Gaussian kernel can be defined as:

$$K(x, y) = \exp(-\gamma \|x - y\|^2)$$

where x and y are the i/p feature vectors, γ is a parameter that controls the width of the Gaussian function, and $\|x - y\|^2$ is the squared Euclidean distance b/w the i/p vectors.

↳ when using a linear kernel in SVM, the decision boundary is a linear hyperplane that separates the different classes in the feature space.

↳ The linear boundary can be useful when the data is already separable by a linear decision boundary or when dealing with high-dimensional data where the use of more complex kernel functions may lead to overfitting.

Polynomial kernel

↳ it is non-linear kernel function that employs polynomial function to transfer the i/p data into the a higher-dimensional feature space.

$$K(x, y) = (x \cdot y + c)^d$$

where x and y are i/p vector feature vectors, c is a constant term, and d is the degree of the polynomial

↳ The constant term is added to, and the dot product of the i/p vector, elevated to the degree of the polynomial.

↳ The decision boundary of an SVM with polynomial kernel might capture more intricate correlations b/w the i/p characteristics because it is a non-linear hyperplane.